Network Security Scanner

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Summary

This script is a command-line network security scanner. It finds live hosts in a target range, scans their ports and services, attempts weak-credential checks (SSH, FTP, Telnet, RDP) using Hydra, and optionally performs deeper vulnerability mapping with Nmap NSE, searchsploit, and Nikto. It saves structured results and produces a detailed markdown report.

Required Tools & Wordlists

Tools & Utilities

nmap, hydra, masscan, git, searchsploit, nikto.

Wordlists

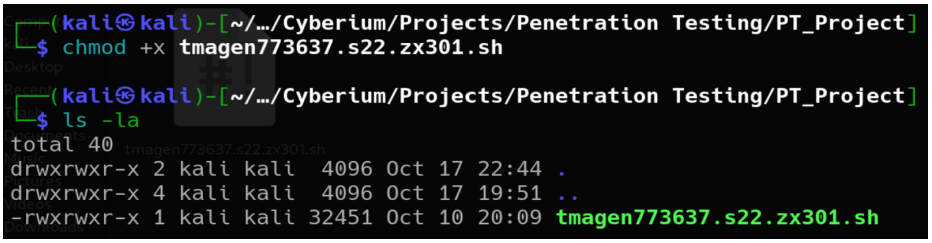
seclists, nmap, rockyou, Hardcoded wordlist

Main Functionalities

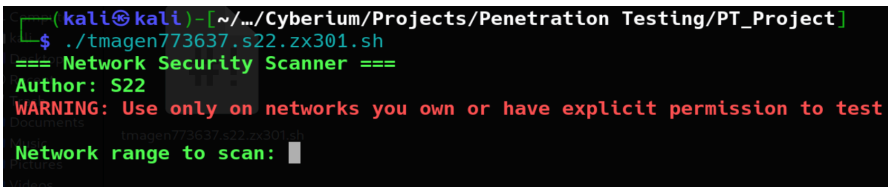
- **Host Discovery:** Uses nmap -sn to find live hosts and filters out local addresses.
- **Port & Service Scanning:** Runs TCP scans (nmap -sT or nmap -sS in full scan) and optionally masscan UDP scans for top ports.
- **Credential Enumeration:** Uses hydra with configurable username/password lists to try logins for SSH (22), FTP (21), Telnet (23), and RDP (3389).
- **Vulnerability Mapping:** Runs Nmap NSE scripts (default,vuln,auth), exports XML for searchsploit, and optionally runs nikto against detected web services.
- **Wordlist Management:** Three choices — Hardcoded wordlist (Internal, No need to download), Download from a GitHub repo, Provide path to desired wordlist.
- **Tool Check & Installer Prompt:** Verifies required tools (nmap, hydra, masscan, git, searchsploit, nikto) and offers to install missing ones via apt utility.
- **Reporting:** Creates a Consolidated_Report.md with host lists, Brute-force findings, Top ports,Searchsploit/Nikto summaries.
- **Post-scan Utilities:** Interactive menu to view a summary, search results, zip output, or print output path.

How to run

- Make the script executable: `chmod +x tmagen773637.s22.zx301.sh`



- Run: `./tmagen773637.s22.zx301.sh`



- Enter target range when prompted (e.g., 192.168.1.0/24).

```
┌──(kali㉿kali)-[~/…/Cyberium/Projects/Penetration Testing/PT_Project]
└─$ ./tmagen773637.s22.zx301.sh
=== Network Security Scanner ===
Author: S22
WARNING: Use only on networks you own or have explicit permission to test

Network range to scan: 192.168.253.0/24
```

- Enter an existing output directory path.

```
┌──(kali㉿kali)-[~/…/Cyberium/Projects/Penetration Testing/PT_Project]
└─$ ./tmagen773637.s22.zx301.sh
=== Network Security Scanner ===
Author: S22
WARNING: Use only on networks you own or have explicit permission to test

Network range to scan: 192.168.253.0/24
Choose path for the output files (must exist): ~/Desktop/
Directory exists: /home/kali/Desktop/
```

- Choose Basic or Full scan and follow menus.

```
┌──(kali㉿kali)-[~/…/Cyberium/Projects/Penetration Testing/PT_Project]
└─$ ./tmagen773637.s22.zx301.sh
=== Network Security Scanner ===
Author: S22
WARNING: Use only on networks you own or have explicit permission to test

Network range to scan: 192.168.253.0/24
Choose path for the output files (must exist): ~/Desktop/
Directory exists: /home/kali/Desktop/

Select scan type:
1) Basic Scan
2) Full Scan (NSE + Vulnerbility mapping + Weak credentials)
3) Exit
Enter choice [1-3]:
```

Use cases

- Classroom labs and safe practice on lab networks.
- Internal red-team / blue-team exercises with permission.
- Quick reconnaissance of allowed test ranges.